

CSCI-338 · THEORY OF COMPUTATION

Mathematical Preliminaries

Sets · Multisets · Set Operations · Sequences



A review of the mathematical background — Sipser, Chapter 0



Sets

A set is an unordered collection of distinct objects.

EXAMPLES

$V = \{ A, E, I, O, U \}$

$\text{bits} = \{ 0, 1 \}$

$\mathbb{Z} = \{ \dots, -2, -1, 0, 1, 2, \dots \}$

CARDINALITY $| S |$

$| V | = 5$ $| \text{bits} | = 2$

Counts how many elements the set has.

MEMBERSHIP $\in / \notin$

$A \in V$ A is an element of V

$B \notin V$ B is not in V



Subsets

$A \subseteq B$ when every element of A also appears in B.

EXAMPLE

$$A = \{ 2, 5 \}$$

$$B = \{ 1, 2, 3, 4, 5 \}$$

$$A \subseteq B \quad \checkmark$$

$$B \subseteq A \quad \times$$

NOT A SUBSET

$$D = \{ 1, 6 \}$$

$$D \not\subseteq B \quad \times$$

because $6 \notin B$

$A \not\subseteq B$ denotes "A is not a subset of B."

P The Power Set

$P(A)$ is the set of all subsets of A .

EXAMPLE · $A = \{ 1, 2, 3 \}$

$$P(A) = \{ \emptyset, \{1\}, \{2\}, \{3\}, \\ \{1,2\}, \{1,3\}, \{2,3\}, \\ \{1,2,3\} \}$$

SUBSET COUNT

$$2^n$$

$$| P(A) | = 8 = 2^3$$

Key rule: a set with n elements has exactly 2^n subsets.



Proper Subsets

$A \subset B$ when $A \subseteq B$ and $A \neq B$ (also written as $A \subsetneq B$)

EXAMPLE

$$A = \{1, 2\}$$

$$B = \{1, 2, 3, 4\}$$

$$A \subsetneq B \quad \checkmark$$

Since $A \subseteq B$ and $A \neq B$



COMMON PITFALL

Don't define it as $|A| \neq |B|$ — that test fails for infinite sets.
The $EVENs \subseteq \mathbb{Z}$, yet they have the **same** cardinality — the map $n \mapsto 2n$ is a bijection.

= Proving Set Equality

MUTUAL CONTAINMENT

$$(X \subseteq Y) \wedge (Y \subseteq X) \Rightarrow X = Y$$

To show two sets are equal, show each contains the other.

THE SAME PATTERN AS $\leq$ ON THE REALS

$$\text{For } a, b \in \mathbb{R}: (a \leq b) \wedge (b \leq a) \Rightarrow a = b$$

This property is called antisymmetry.



Multisets

A **multiset** is a set that allows duplicate elements. e.g. $M = \{ 1, 1, 2, 3, 4, 4, 5 \}$

FORMAL DEFINITION

A multiset is an ordered pair (A, m) where:

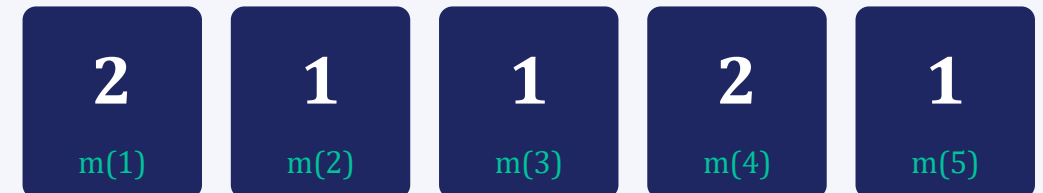
A — the underlying set

$m : A \rightarrow \mathbb{Z}^+$ — the multiplicity function
(copies of each element)

EXAMPLE

$M = \{ 1, 2, 1, 3, 4, 5, 4 \}$

$A = \{ 1, 2, 3, 4, 5 \}$





Important Sets

 $\mathbb{Z}$

Integers

$$\mathbb{Z} = \{ \dots, -2, -1, 0, 1, 2, \dots \}$$

 $\mathbb{Z}^{\geq 0}$

Non-negative integers

$$\mathbb{Z}^{\geq 0} = \{ 0, 1, 2, 3, \dots \}$$

 $\mathbb{N}$

Natural numbers

$$\mathbb{N} = \{ 1, 2, 3, \dots \}$$

 $\mathbb{Q}$

Rational numbers

$$\mathbb{Q} = \left\{ \frac{p}{q} : p, q \in \mathbb{Z}, q \neq 0 \right\}$$

 $\mathbb{Z}^+$

Positive integers

$$\mathbb{Z}^+ = \{ 1, 2, 3, 4, \dots \}$$

 $\mathbb{R}$

Real numbers

all points on the number line

 $\mathbb{Z}^-$

Negative integers

$$\mathbb{Z}^- = \{ \dots, -3, -2, -1 \}$$

$$\text{Note: } \mathbb{Z} = \mathbb{Z}^+ \cup \mathbb{Z}^- \cup \{ 0 \}$$



Set-Builder Notation

$$S = \{ \mathbf{x} : \mathbf{rule\ about\ x} \}$$

“the set of all x such that the rule holds” — read the colon ($:$) as “such that”

RATIONAL NUMBERS

$$\mathbb{Q} = \{ a : p, q \in \mathbb{Z} \wedge q \neq 0 \wedge a = p/q \}$$

every a that equals a fraction p/q with nonzero denominator

EVEN INTEGERS

$$EVENS = \{ e \in \mathbb{Z} : e \bmod 2 = 0 \}$$

every integer e that leaves remainder 0 when divided by 2



Set Operations

$$A = \{1, 3, 5, 6\} \quad B = \{2, 4, 6, 8\} \quad U = \{1, 2, 3, 4, 5, 6, 7, 8\}$$



Union

in A or B

$$A \cup B = \{1, 2, 3, 4, 5, 6, 8\}$$



Intersection

in A and B

$$A \cap B = \{6\}$$



Complement

in U but not A

$$\bar{A} = \{2, 4, 7, 8\}$$



Symmetric difference

in exactly one

$$A \oplus B = \{1, 3, 5, 2, 4, 8\}$$



Set Difference

$A - B$ or $A \setminus B$ the elements that are in A but not in B .

EXAMPLE

$$A = \{1, 3, 5, 6\}$$

$$B = \{2, 4, 6, 8\}$$

$$A - B = \{1, 3, 5\}$$

NOT COMMUTATIVE

$$A - B \neq B - A \text{ (unless } A = B\text{)}$$

In fact, $A - B$ and $B - A$ are always **disjoint** — so if they were equal, both would be empty, forcing $A = B$.

Handy identity: $A \cup \bar{A} = U$ — a set together with its complement fills the whole universe.



Sequences & Tuples

A **sequence** is an ordered list of elements — unlike a set, order matters and repeats are allowed.

ORDER MATTERS

$\langle 1, 5, 6, 7 \rangle = \langle 1, 5, 6, 7 \rangle$

$\langle 1, 5, 6, 7 \rangle \neq \langle 1, 6, 5, 7 \rangle$

TUPLES

A finite sequence is a **tuple**.

A 2-tuple is a **pair**: (a, b)

WHY IT MATTERS IN THIS COURSE

A graph is a tuple $G = (V, E)$ — V is the set of vertices, E is the set of edges. (Automata will be tuples too.)

Key Takeaways

- 1 Subset vs. proper subset.** $A \subseteq B$ means every element of A is in B ; $A \subsetneq B$ adds $A \neq B$ — never test it by cardinality.
- 2 Count subsets with powers of two.** A set of n elements has exactly 2^n subsets — that is $|P(A)|$.
- 3 Prove equality by mutual containment.** $X \subseteq Y$ and $Y \subseteq X$ together give $X = Y$ — the same antisymmetry as $\leq$ on $\mathbb{R}$.
- 4 Know your operations.** $\cup$ (or), $\cap$ (and), complement (not), $\oplus$ (exactly one), and $-$ which is not commutative.
- 5 Order distinguishes sequences.** Tuples are ordered and can repeat — graphs $G = (V, E)$, and later automata, are tuples.

Let's do a simple exercise

- Let's draw
 - $A \cup B$
 - $A \cap B$
 - $A \oplus B$
 - $A - B$
 - $B - A$ in Venn diagrams.